DERIVATIVE OF A FUNCTION

Chapter 2
Derivative of a Function

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2.1 Formal Definition of the Derivative

Definition 2.1 If f is a function of an independent variable, say x, then the **derivative** of f at x_0, denoted by $f'(x_0)$, is given by

$$f'(x_0) = \lim_{x \mapsto 0} \frac{f(x_0 + h) - f(x_0)}{h},$$

if this limit exists. If $f'(x_0)$ exists, then f is said to be **differentiable at** x_0. The function f is said to be **differentiable** if it is differentiable at each point in the domain of f.

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From Definition 2.1, a systematic procedure can be immediately formulated for obtaining the derivative of f at x, if it exists.

This is called the **three-step rule**:

1. Simplify $f(x+h) - f(x)$.
2. Divide $f(x+h) - f(x)$ by $h \neq 0$, that is, solve for $\dfrac{f(x+h) - f(x)}{h}$.
3. Evaluate $\lim\limits_{h \mapsto 0} \dfrac{f(x+h) - f(x)}{h}$. By definition, this limit, if it exist, is $f'(x)$.

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Example 2.2 Use the 3-Step Rule to find the derivative $f'(x)$ of the function $f(x) = 4x - 5$.

Solution: Following the 3-Step Rule:

1. $f(x+h) - f(x) = [4(x+h) - 5] - (4x - 5) = 4x + 4h - 5 - 4x + 5 = 4h.$
2. $\dfrac{f(x+h) - f(x)}{h} = \dfrac{4h}{h} = 4.$
3. $\lim\limits_{h \mapsto 0} \dfrac{f(x+h) - f(x)}{h} = \lim\limits_{h \mapsto 0} 4 = 4.$

Therefore, $f'(x) = 4$.

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Example 2.3 Use the 3-Step Rule to find the derivative $f'(x)$ of the function $f(x) = 3x^2 - x$.

Solution: Following the 3-Step Rule:

1. $$\begin{aligned} f(x+h) - f(x) &= [3(x+h)^2 - (x+h)] - (3x^2 - x) \\ &= 3(x^2 + 2xh + h^2) - x - h - 3x^2 + x \\ &= 3x^2 + 6xh + 3h^2 - h - 3x^2 \\ &= 6xh + 3h^2 - h. \end{aligned}$$

2. $$\frac{f(x+h) - f(x)}{h} = \frac{6xh + 3h^2 - h}{h} = \frac{h(6x + 3h - 1}{h} = 6x + 3h - 1.$$

3. $$\lim_{h \mapsto 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \mapsto 0} 6x + 3h - 1 = 6x + 3(0) - 1 = 6x - 1.$$

Therefore, $f'(x) = 6x - 1$.

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Example 2.4 Use the 3-Step Rule to find the derivative $f'(x)$ of the function $f(x) = \frac{1}{x}, x \neq 0$.

Solution: Following the 3-Step Rule:

1. $f(x+h) - f(x) = \dfrac{1}{x+h} - \dfrac{1}{x} = \dfrac{x-(x+h)}{x(x+h)} = \dfrac{-h}{x(x+h)}$.

2. $\dfrac{f(x+h)-f(x)}{h} = \dfrac{\frac{-h}{x(x+h)}}{h} = \frac{-h}{x(x+h)} \cdot \frac{1}{h} = \frac{-1}{x(x+h)}$.

3. $\lim\limits_{h \mapsto 0} \dfrac{f(x+h)-f(x)}{h} = \lim\limits_{h \mapsto 0} \frac{-1}{x(x+h)} = \frac{-1}{x(x+0)} = -\frac{1}{x^2}$.

Therefore, $f'(x) = -\frac{1}{x^2}$.

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Example 2.5 Use the 3-Step Rule to find the derivative $f'(x)$ of the function $f(x) = \dfrac{1}{\sqrt{x}}$ for $x > 0$.

Solution: Following the 3-Step Rule:

1. $$f(x+h) - f(x) = \frac{1}{\sqrt{x+h}} - \frac{1}{\sqrt{x}} = \frac{\sqrt{x} - \sqrt{x+h}}{\sqrt{x+h}\sqrt{x}} \cdot \frac{\sqrt{x} + \sqrt{x+h}}{\sqrt{x} + \sqrt{x+h}}$$
$$= \frac{x - (x+h)}{\sqrt{x+h}\sqrt{x}(\sqrt{x} + \sqrt{x+h})}$$
$$= \frac{-h}{\sqrt{x+h}\sqrt{x}(\sqrt{x} + \sqrt{x+h})}.$$

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2. $$\frac{f(x+h)-f(x)}{h} = \frac{\frac{-h}{\sqrt{x+h}\sqrt{x}(\sqrt{x}+\sqrt{x+h})}}{h}$$

$$= \frac{-h}{\sqrt{x+h}\sqrt{x}(\sqrt{x}+\sqrt{x+h})} \cdot \frac{1}{h}$$

$$= \frac{-1}{\sqrt{x+h}\sqrt{x}(\sqrt{x}+\sqrt{x+h})}.$$

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3. $$\lim_{h\mapsto 0}\frac{f(x+h)-f(x)}{h} = \lim_{h\mapsto 0}\frac{-1}{\sqrt{x+h}\sqrt{x}(\sqrt{x}+\sqrt{x+h})}$$
$$= \frac{-1}{\sqrt{x+0}\sqrt{x}(\sqrt{x}+\sqrt{x+0})}$$
$$= \frac{-1}{\sqrt{x}\sqrt{x}(2\sqrt{x})} = \frac{-1}{2(\sqrt{x})^3}.$$

Therefore, $f'(x) = \dfrac{-1}{2\sqrt{x^3}}$.

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Exercise 2.6 Use the 3-Step Rule to find the derivative $f'(x)$ of the following functions.

1. $f(x) = 7x - 2$
2. $f(x) = 5$
3. $f(x) = -\frac{3}{x}, x \neq 0$
4. $f(x) = \sqrt{x+3}$
5. $f(x) = x^2 + 1$
6. $f(x) = x^3 - 9$
7. $f(x) = \sqrt{2x-3}$
8. $f(x) = \frac{2}{x+2}, x \neq -2$
9. $f(x) = \frac{x}{x+2}$
10. $f(x) = \frac{x}{2x-5}$

2.2 Theorems on Differentiation

Definition 2.7 The process of finding the derivative of a given function is called **differentiation**.

Notations If the function f is defined by the equation $y = f(x)$, then the derivative of f at x can be denoted by any of the following symbols:

$$f'(x), \quad \frac{dy}{dx}, \quad D_x f, \quad D_x f(x), \quad D_x y, \quad \frac{d}{dx}[f(x)], \quad y'$$

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Theorem 2.8 [Derivative of a constant]
If c is any constant and for all x $f(x) = c$, then $f'(x) = 0$.

Theorem 2.9 [Derivative of a power]
If r is a real number and $f(x) = x^r$, then $f'(x) = rx^{r-1}$.

Illustrations:

1. $f(x) = \sqrt{2}$, we have $f'(x) = 0$
2. $f(x) = x^3$, we have $f'(x) = 3x^2$
3. If $g(x) = \ln \sqrt[3]{7}$, then $g'(x) = 0$

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$h(x) = \sqrt{x}$

Solution: Rewriting the function, $h(x) = \sqrt{x} = x^{\frac{1}{2}}$.

Then for $r = \frac{1}{2}$, we have $h'(x) = \frac{1}{2}x^{\frac{1}{2}-1} = \frac{1}{2}x^{-\frac{1}{2}} = \frac{1}{2x^{\frac{1}{2}}} = \frac{1}{2\sqrt{x}}$

5. $f(x) = \frac{1}{x^5}$

Solution: Rewriting the function, $f(x) = \frac{1}{x^5} = x^{-5}$.

Then for $r = -5$, $f'(x) = -5x^{-5-1} = -5x^{-6} = \frac{-5}{x^6}$

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Theorem 2.10 If h is a differentiable function at x and

$$f(x) = c \cdot h(x),$$

where c is a constant, then f is differentiable at x and

$$f'(x) = c \cdot h'(x).$$

Illustration:

1. $f(x) = 6x^4$

Solution: $f'(x) = 6 \cdot 4x^3 = 24x^3$

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2. $f(x) = \dfrac{4}{\sqrt{x}}$

Solution: Note that $f(x) = \dfrac{4}{\sqrt{x}} = \dfrac{4}{x^{\frac{1}{2}}} = 4x^{-\frac{1}{2}}$ so that

$$f'(x) = 4\left(-\tfrac{1}{2}x^{-\frac{1}{2}-1}\right) = -2x^{\frac{-1-2}{2}} = -2x^{\frac{-3}{2}} = -\frac{2}{x^{\frac{3}{2}}} = \frac{-2}{\sqrt{x^3}}.$$

Theorem 2.11 [*Derivative of a Sum*]
g and h are differentiable at x and

$$f(x) = g(x) + h(x),$$

then f is differentiable at x and

$$f'(x) = g'(x) + h'(x).$$

Illustration:

1. $f(x) = 3x^5 - 7x^3$

Solution: $f'(x) = 15x^4 - 21x^2$

2. $f(x) = 2x^5 + 3x^4 - 6x^3 + 5x^2 - 5x - 8$

Solution: $f'(x) = 10x^4 + 12x^3 - 18x^2 + 10x - 5$

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Theorem 2.12 [*Derivative of a Product*]
If g and h are differentiable at x and

$$f(x) = g(x) \cdot h(x),$$

then f is differentiable at x and

$$f'(x) = g(x) \cdot h'(x) + h(x) \cdot g'(x).$$

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Illustration:

$f(x) = (3x^2 + 2x)(x^2 - 5x + 1)$

Solution:

$$
\begin{aligned}
f'(x) &= (3x^2 + 2x)(2x - 5) + (x^2 - 5x + 1)(6x + 2) \\
&= 6x^3 - 15x^2 + 4x^2 - 10x + 6x^3 + 2x^2 - 30x^2 - 10x + 6x + 2 \\
&= -39x^2 - 14x + 2
\end{aligned}
$$

Theorem 2.13 [*Derivative of a Quotient*]
If g and h are differentiable at x and

$$f(x) = \frac{g(x)}{h(x)}$$

with $h(x) \neq 0$, then f is differentiable at x and

$$f'(x) = \frac{h(x)g'(x) - g(x)h'(x)}{[h(x)]^2}.$$

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Illustration:

1. $f(x) = \frac{x^2+1}{3x-1}$

Solution:

$$f'(x) = \frac{(3x-1)(2x)-(x^2+1)(3)}{(x-1)^2}$$

$$= \frac{6x^2-2x-3x^2-3}{(x-1)^2}$$

$$= \frac{3x^2-2x-3}{(x-1)^2}$$

Theorem 2.14 [*Derivative of a Composite Function - The Chain Rule*] Suppose that f, g, and u are functions with

$$f(x) = g(u(x))$$

and suppose that g and u are differentiable at x. Then f is differentiable at x and

$$f'(x) = g'(u(x)) \cdot u'(x).$$

Corollary 2.15 If

$$f(x) = [u(x)]^r,$$

where r is any real number, and u is a differentiable function, then

$$f'(x) = r[u(x)]^{r-1}u'(x).$$

Illustration:

1. $f(x) = (x^2 + 3x - 2)^5$

Solution: $f'(x) = 5(x^2 + 3x - 2)^4(2x + 3)$

Example 2.16 Find $\dfrac{dy}{dx}$ given that $y = \dfrac{3}{x^3 - 4x^2 - 5x + 2}$.

Solution: Note that $y = \dfrac{3}{x^3 - 4x^2 - 5x + 2}$ can be written as

$$y = 3(x^3 - 4x^2 - 5x + 2)^{-1}.$$

Applying Corollary 2.15 with $n = -1$, we obtain

$$\begin{aligned}\frac{dy}{dx} &= 3(-1)(x^3 - 4x^2 - 5x + 2)^{-2}(3x^2 - 8x - 5)\\ &= \frac{-3(3x^2 - 8x - 5)}{(x^3 - 4x^2 - 5x + 2)^2}\\ &= \frac{-9x^2 + 24x + 15}{(x^3 - 4x^2 - 5x + 2)^2}.\end{aligned}$$

This example could also be solved using Theorem 2.13. That is if

$$y = \frac{3}{x^3 - 4x^2 - 5x + 2}, \text{ then}$$

$$\frac{dy}{dx} = \frac{(x^3 - 4x^2 - 5x + 2)(0) - 3(3x^2 - 8x - 5)}{(x^3 - 4x^2 - 5x + 2)^2}$$

$$= \frac{-9x^2 + 24x + 15}{(x^3 - 4x^2 - 5x + 2)^2}.$$

Example 2.17 Given the function $f(x) = \left(\frac{5x+2}{2x-3}\right)^5$, find $f'(x)$.

Solution:

This example requires a combination of Corollary 2.15 and Theorem 2.13. Thus,

$$\begin{aligned} f'(x) &= 5\left(\frac{5x+2}{2x-3}\right)^4\left(\frac{(2x-3)(5)-(5x+2)(2)}{(2x-3)^2}\right) \\ &= 5\left(\frac{5x+2}{2x-3}\right)^4\left(\frac{10x-15-10x-4}{(2x-3)^2}\right) \\ &= 5\left(\frac{5x+2}{2x-3}\right)^4\frac{-19}{(2x-3)^2} \\ &= \frac{-95(5x+2)^4}{(2x-3)^6}. \end{aligned}$$

Example 2.18 Find $\frac{dy}{dx}$ given the function $y = (x^2 - 3x + 1)^{12}(2x - 3)^{15}$.

Solution: This example requires a combination of Corollary 2.15 and Theorem 2.12. That is,

$$\begin{aligned}
\frac{dy}{dx} &= 30(x^2 - 3x + 1)^{12}(2x - 3)^{14} + (2x - 3)^{15}12(x^2 - 3x + 1)^{11}(2x - 3) \\
&= 6(x^2 - 3x + 1)^{11}(2x - 3)^{14}\left[5(x^2 - 3x + 1) + 2(2x - 3)^2\right] \\
&= 6(x^2 - 3x + 1)^{11}(2x - 3)^{14}\left[5x^2 - 15x + 5 + 2(4x^2 - 12x + 9)\right] \\
&= 6(x^2 - 3x + 1)^{11}(2x - 3)^{14}(5x^2 - 15x + 5 + 8x^2 - 24x + 18) \\
&= 6(x^2 - 3x + 1)^{11}(2x - 3)^{14}(13x^2 - 39x + 23)
\end{aligned}$$

Exercise 2.19 Find the derivative of each of the following:

1. $f(x) = 8 - x^3$
2. $f(x) = 3x^3 - 12x + 5$
3. $f(x) = 3x\sqrt[3]{2x-4}$
4. $f(x) = (5x^2 - 1)(7x^3 + 3)$
5. $f(x) = \dfrac{x^3 - 1}{x^3 + 1}$
6. $f(x) = \dfrac{2x+1}{x+5}(3x - 1)$
7. $y = (3x - 2)^{-4}$
8. $y = (x^3 + 7x + 2 - x^{-}3)^5$
9. $d\left[(2x-1)^4(3x+2)^5\right]$
10. $g(x) = \frac{(2x-3)^3}{(3x-2)^4}$
11. $h(x) = \left(\frac{x-1}{x+1}\right)^{15}$
12. $D\left[\frac{1}{\sqrt[3]{x^2+x+1}}\right]$